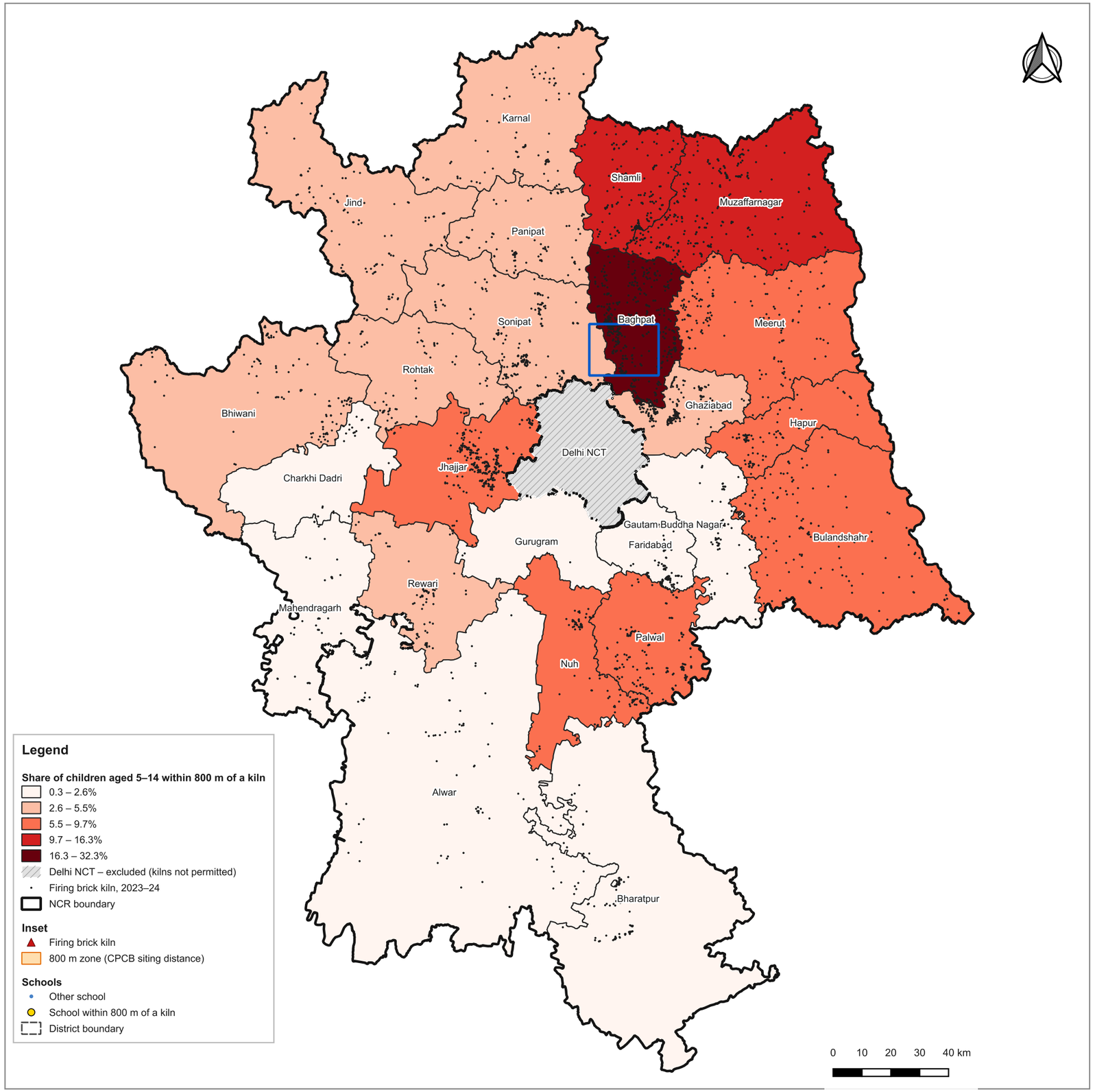
One in three children in Baghpat lives within 800 m of a firing brick kiln

Brick kilns found by AI on Sentinel-2 imagery, and the children and schools inside the CPCB 800 m siting distance · National Capital Region, 2023–24 firing season

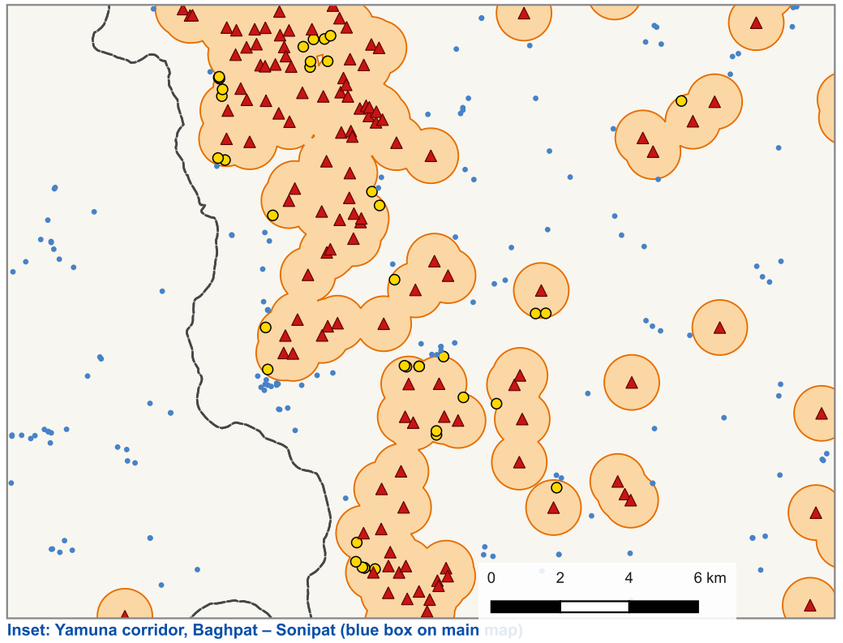


Legend

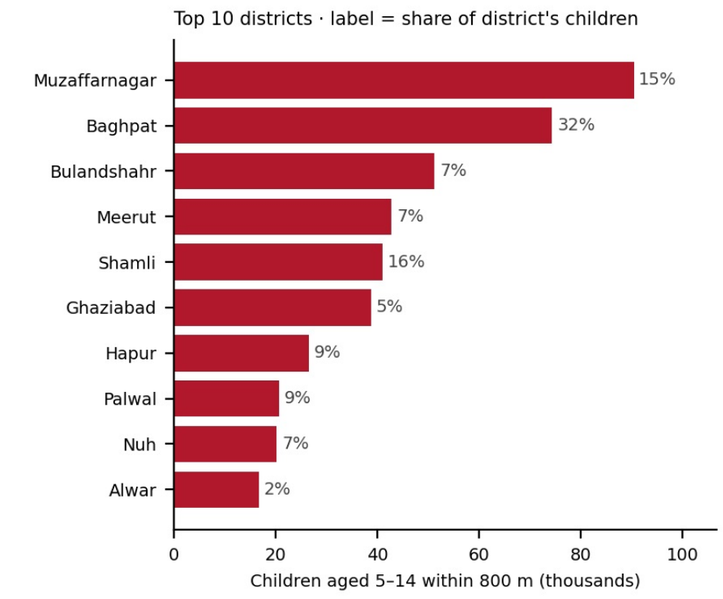
Share of children aged 5–14 within 800 m of a kiln
- 0.3 – 2.6%
- 2.6 – 5.5%
- 5.5 – 9.7%
- 9.7 – 16.3%
- 16.3 – 32.3%
- Delhi NCT – excluded (kilns not permitted)
- Firing brick kiln, 2023–24
- NCR boundary

Inset
- Firing brick kiln
- 800 m zone (CPCB siting distance)

Schools
- Other school
- School within 800 m of a kiln
- District boundary



Inset: Yamuna corridor, Baghpat – Sonipat (blue box on main map)



Key numbers (NCR excluding Delhi)

≈ 528,000 children aged 5–14 (6.4%) live within 800 m of a firing kiln
835 schools stand within 800 m · 23% of 3,057 kilns have a school within 800 m
Kilns near schools were no more likely to stop firing by 2025–26 (10.2% vs 10.7%)

Method: YOLO11 oriented-box detector on Sentinel-2 winter composites, filtered by land cover, verified by SWIR hot-spot firing frequency, non-kiln heat sources removed; every kiln checked by eye, doubtful ones by R. Sinha (208 non-kilns removed). Children: WorldPop 2020, ages 5–14. Schools: UDISE (DataMeet), location-checked, shared points removed.
Data: Copernicus Sentinel-2 (ESA), SentinelKilnDB (IIT Gandhinagar, CC BY-NC 4.0), ESA WorldCover, WorldPop, UDISE/DataMeet, OpenStreetMap contributors, NCRPB boundaries (rebuilt).
CRS: WGS 84 / UTM zone 43N (EPSG:32643) · Produced 20 October 2026 · KilnWatch – Rashmi Sinha